SUNWAY UNIVERSITY

Jeffrey Cheah Foundation
Nurturing the Seeds of Wisdom

Practical 04: Setting up a NFS Server

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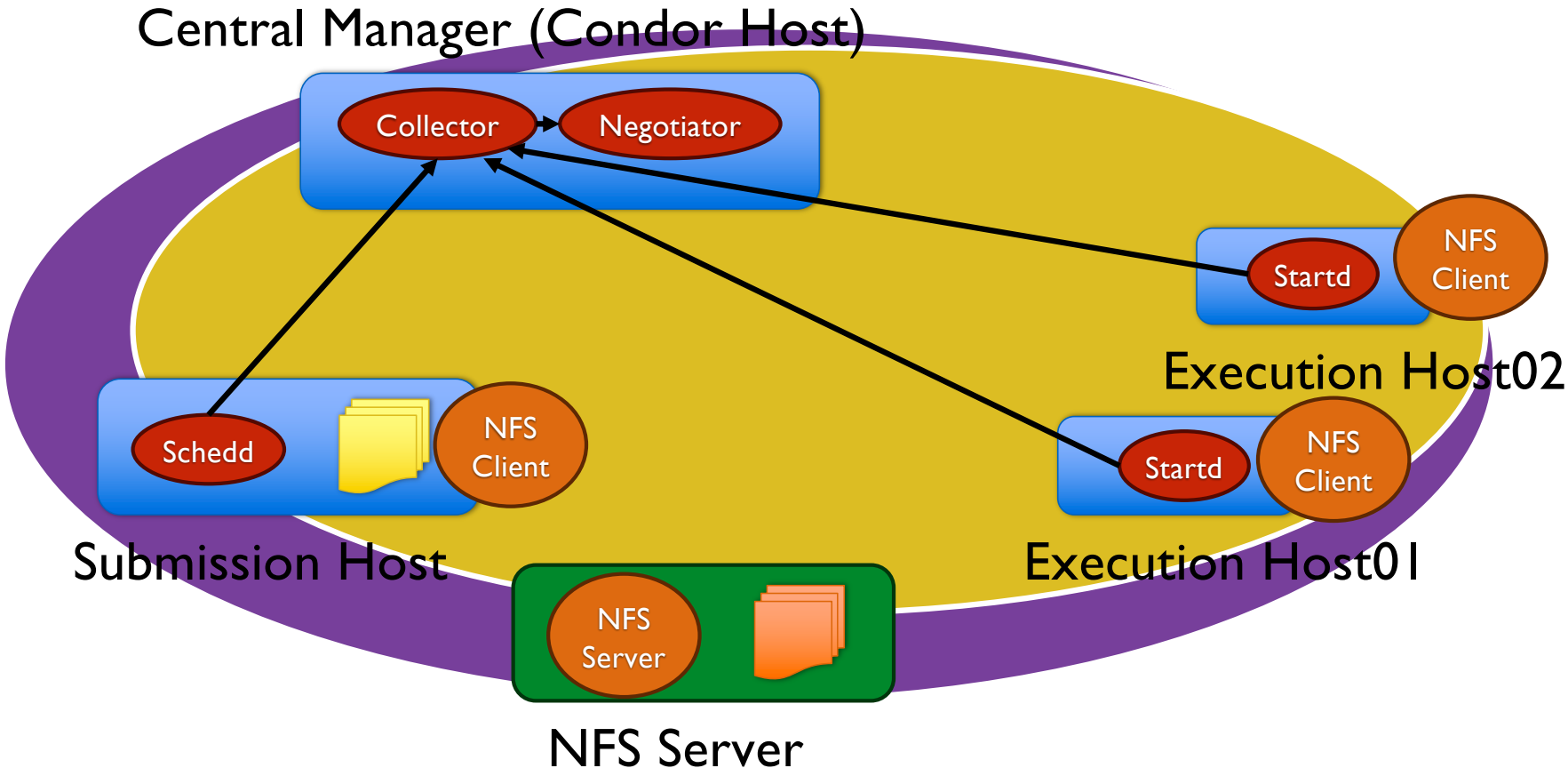
NET3204
Distributed Systems

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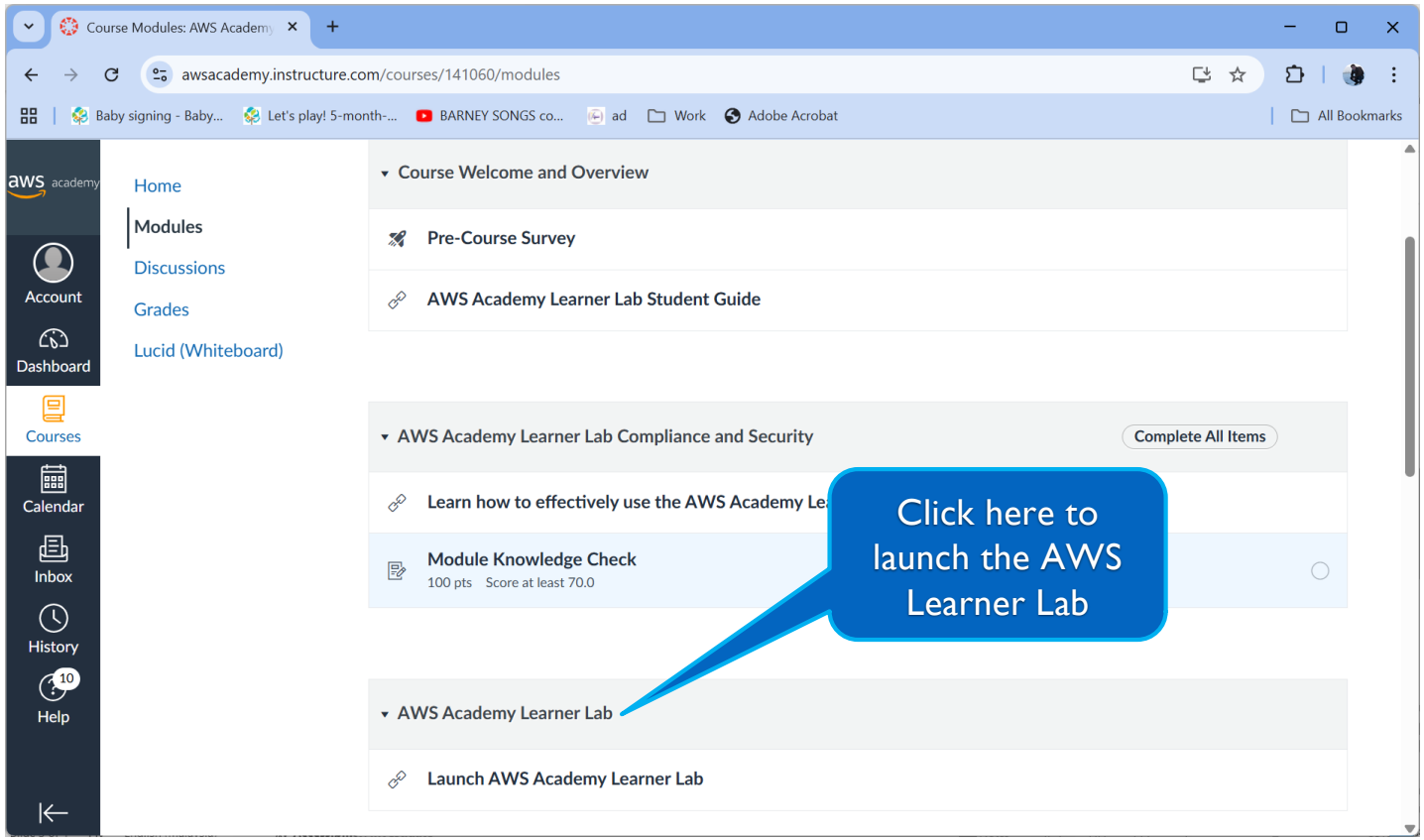
Practical Description

- In this practical, you will learn how to:
  - Sign in to the AWS Academy
  - Launch ONE virtual machine from the AWS EC2
  - Set up a NFS server.
  - Set up NFS client on the Submission Host and 2 Execution Hosts of your HTCondor pool
  - Create a simple shared directory in Ubuntu home.
  - Create a shared directory for HTCondor job submisison.

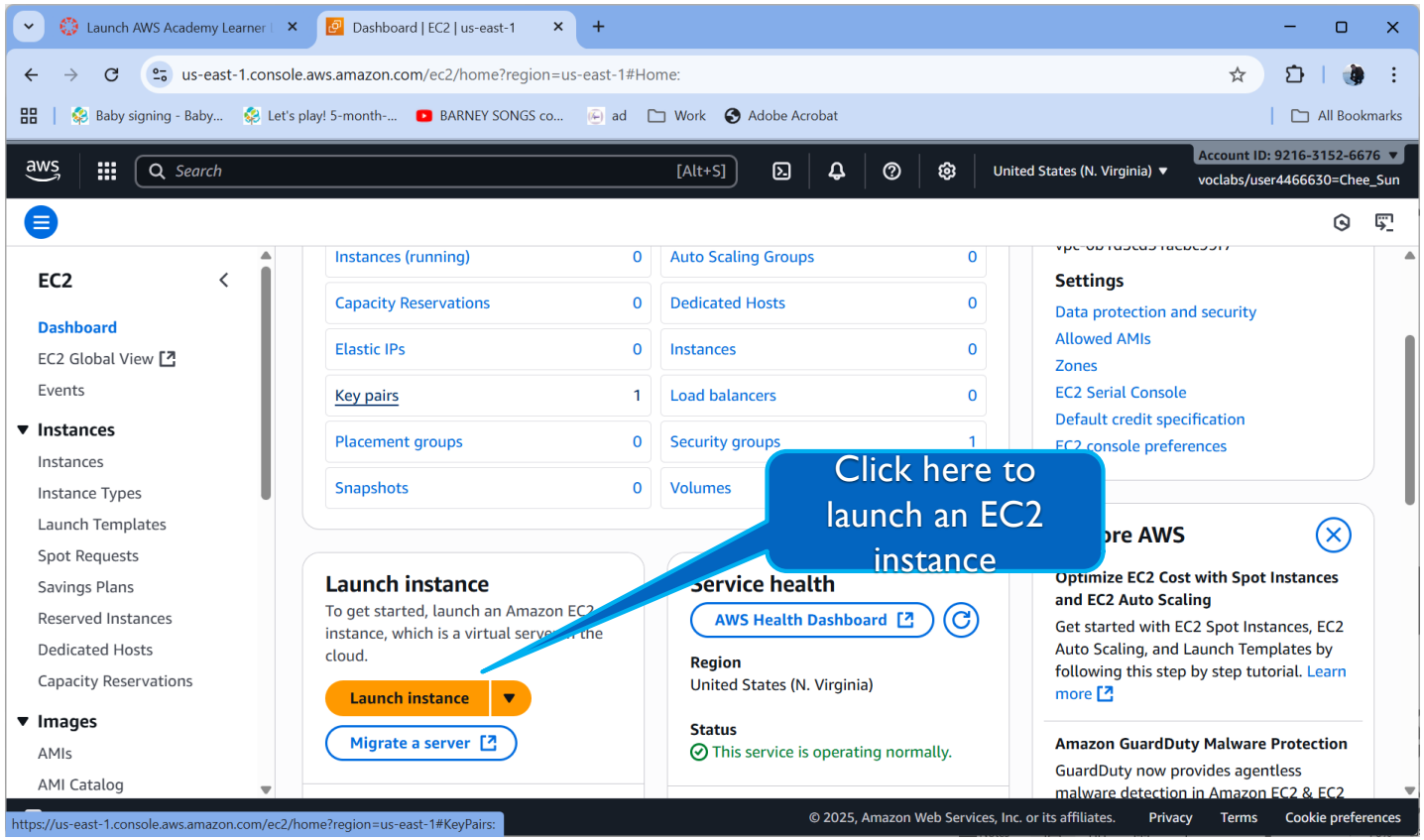
Our deployment plan today



1. Sign in to the Course Page

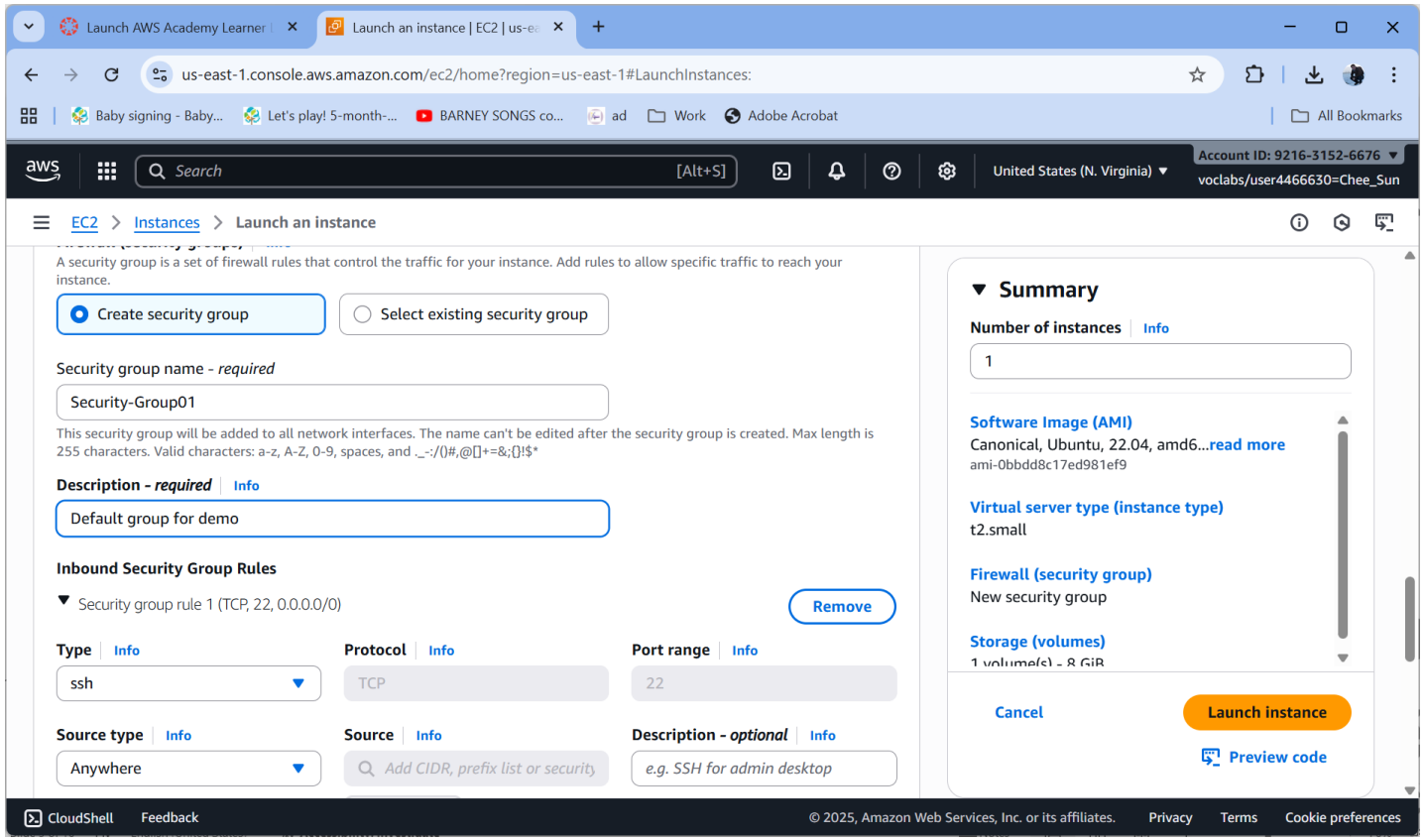


2. Launch ONE EC2 instance



You need to launch 1 instances, following the steps you used in Practical02

3. Add into the Condor Pool security group



You can use back the same security group created in Practical03. Make sure your instance is using the same security group as your Condor Pool.

4. Install NFS server on new server

- Follow the instruction at https://documentation.ubuntu.com/server/how-to/networking/install-nfs/ to set up NFS Sever
- Install NFS server: **sudo apt install nfs-kernel-server**
- Start the server: **sudo systemctl start nfs-kernel-server.service**
- Create a shared directory at ubuntu home directory named storage: **mkdir /home/ubuntu/storage**
- Configure the directories to be exported by adding them to the /etc/exports file: **/home/ubuntu/storage *(rw,sync,subtree_check)**
- Apply the new configuration: **sudo exportfs –a**
- Show the exported directories: **showmount -e <*NFSServerIP*>**

5. Install NFS Client on condor pool

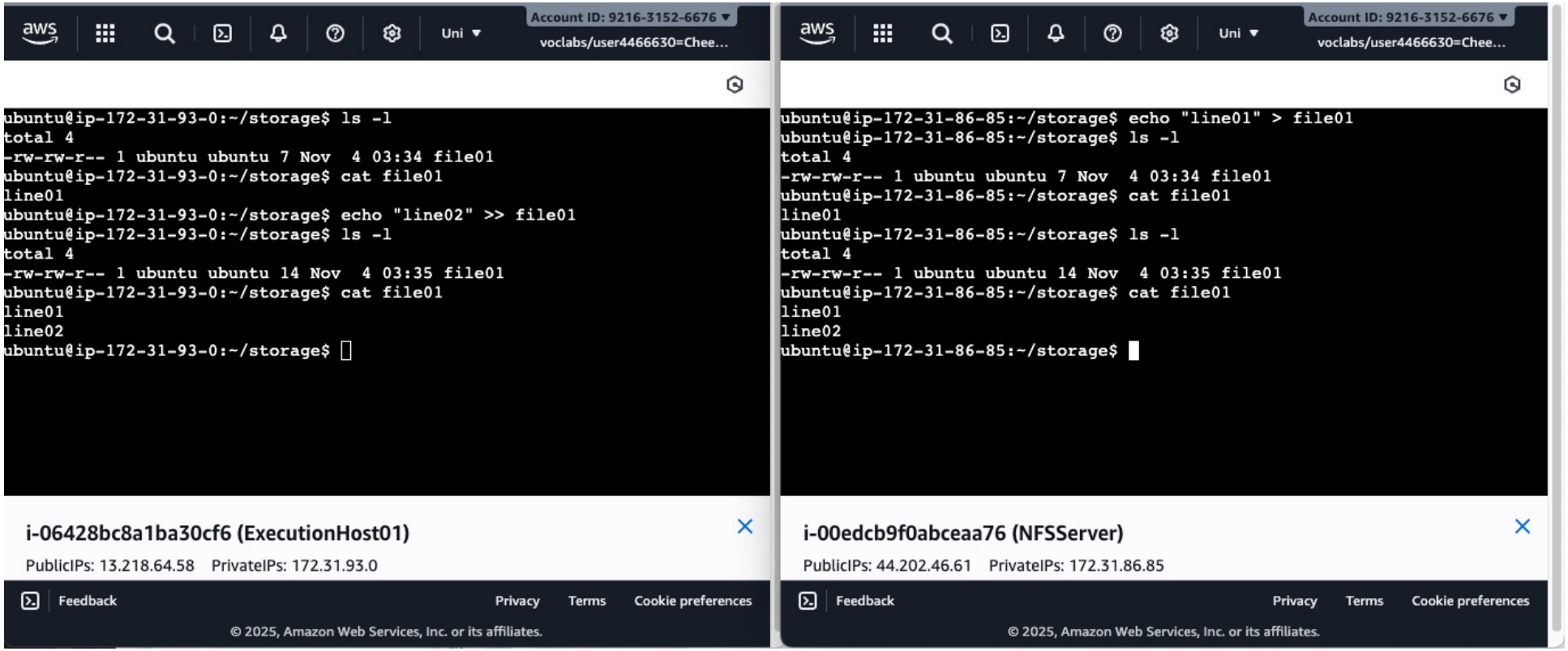
- For all of the machines in condor pool, install NFS Client:
  sudo apt install nfs-common
- In **ONE** of the execution host, create a storage directory in ubuntu home as the mounting point: **mkdir /home/ubuntu/storage**
- Mount the NFS server directory into this local directory: **sudo mount *<NFSServerIP>*:/home/ubuntu/storage /home/ubuntu/storage**

  Remark:
  *1. replace **<NFSServerIP>** with the IP address of your NFS Server, or the hostname (if you have added into your /etc/hosts).*
  2. This is just simple demo for NFS. Thus, only configure in one of the condor nodes.

6. Test the deployment

- On NFS server:
  - Create a new file **file01** in the **/home/ubuntu/storage** directory, with a sentence **line01**
- On the client machine:
  - Listing the directory **/home/ubuntu/storage**. The **file01** shall appear.
  - Modify the file01 by adding **line02**.
- Back on NFS server:
  - Show the content of **file01**.

Output of step 6



Is it sufficient for HTCondor Pool?

- NO!!!
- Reason:
  - The folder you are creating is in the home directory of **ubuntu** user, but HTCondor is executing the job as **condor** user.
  - Thus, it won't be able to access your folder in **ubuntu** home, unless you change the mode to **777** or you add **condor** user into **ubuntu group** ☺

7. Create shared directory for Condor pool in NFS File Server

Let's create a proper shared directory for HTCondor pool.

ONLY in the file server, perform the following:

- Create a shared directory at /srv named storage: **sudo mkdir /srv/data**
- Change the owner of the directory: **sudo chown ubuntu:ubuntu /srv/data**
- Change the mode of the directory: **sudo chmod 775 /srv/data/**
- Configure the directories to be exported by adding them to the /etc/exports file: **/srv/data *(rw,sync,no_subtree_check)**
- Apply the new configuration: **sudo exportfs -a**

8. Add new user & change directory setting

ONLY in the file server, perform the following:

- Add new group: **sudo groupadd condor**
- Add new user: **sudo useradd -m -g condor condor**
- Add **condor** user into **ubuntu** group: **sudo usermod -aG ubuntu condor**
- Check condor user status: **id condor**
  the output is:
  id=1001(condor) gid=1001(condor) groups=1001(condor),1000(ubuntu)

9. Create directory on ExecutionHost and SubmissionHost

In all of the ExecutionHost and SubmissionHost, perform the following:

- Create new **data** directory in **/mnt**: **sudo mkdir /mnt/data**
- Change owner to **ubuntu**: **sudo chown ubuntu:ubuntu /mnt/data**
- Change the access mod of the folder: **chmod 775 /mnt/data/**
- Mount the server directory **/srv/data** to **/mnt/data**: **sudo mount <*NFSServerIP*>:/srv/data /mnt/data/**

Now your shared directory for HTCondor Pool is ready!

Putting all together ☺

Running a Python job on Condor Pool with NFS Shared File System

(Source:adapted from https://cluster.cs.wwu.edu/examples/python.html)

1. Create a simple Python script

- Create a file named **test_script.py** as below:

```
#!/usr/bin/env python3

import getpass, os, sys

print('Hello, ', getpass.getuser(), '!', sep='')
print("You're running on the host", os.uname()[1])
print("The process arguments are:")
for arg in range(len(sys.argv)):
  print('argv[{}]:'.format(arg), sys.argv[arg])
```

- Make it executable: **chmod +x test_script.py**

2. Create a submission script

- Create a file named **test_script.sub** as below:

```
Universe   = vanilla
Executable = test_script.py
Arguments  = -arg1 -arg2 -arg3

Output     = test_script.out
Error      = test_script.err
Log        = test_script.log

Request_Cpus   = 1
Request_Memory = 2GB

Queue
```

3. Test the script locally and in Condor Pool as a Python job

- Locally:

```
ubuntu@ip-172-31-92-248:~$ ./test_script.py
Hello, ubuntu!
You're running on the host ip-172-31-92-248
The process arguments are:
argv[0]: ./test_script.py
```

- In Condor Pool:

```
ubuntu@ip-172-31-92-248:~$ condor_submit test_script.sub
Submitting job(s).
1 job(s) submitted to cluster 10.
ubuntu@ip-172-31-92-248:~$ cat test_script.out
Hello, nobody!
You're running on the host ip-172-31-83-177
The process arguments are:
argv[0]: /var/lib/condor/execute/dir_1068/scratch/test_script.py
argv[1]: -arg1
argv[2]: -arg2
argv[3]: -arg3
```

4. Set up Python virtual environment in the shared file system

- Install the **venv** module: **sudo apt install python3.10-venv**
- Create the virtual environment: **python3 -m venv /mnt/data/my_venv**
- Load the new environment: **source /mnt/data/my_venv/bin/activate**

```
ubuntu@ip-172-31-92-248:~$ source /mnt/data/my_venv/bin/activate
(my_venv) ubuntu@ip-172-31-92-248:~$
```

Notes: Python virtual environment allows each user to create own execution environment, so that can prevent incompatibility and libraries dependencies issue. This is very useful for HPC shared pool.

5. Install necessary packages in your environment

- Update the **core** modules: **python3 -m pip install --upgrade pip setuptools wheel**
- Install the Figcow package: **python3 -m pip install figcow**
- Check the install packages: **pip list**

```
(my_venv) ubuntu@ip-172-31-92-248:~$ pip list
Package    Version
---------- -------
figcow     1.0.3
pip        25.3
setuptools 80.9.0
wheel      0.45.1
```

*Notes: Make sure you're performing this installation your own environment **my_venv** (see the prompt **my_venv**).*

6. Create the Figcow Python script

- Create a file named **test_venv.py** as below:

```
#!/usr/bin/env python3

import figcow, getpass, os, sys

print('Hello, ', getpass.getuser(), '!', sep='')
print("You're running on the host", os.uname()[1])

print(figcow.cow(' '.join((sys.argv[1:]))))
```

- Make it executable: **chmod +x test_venv.py**
- Move it to the mounted directory: **cp test_venv.py /mnt/data**

7. Create a wrapper script

- Create a file named **run_py.sh** as below:

```
#!/bin/sh

. /mnt/data/my_venv/bin/activate
exec python "$@"
```

- Make it executable: **chmod +x run_py.sh**

*Notes: This wrapper script will first active your **my_venv** environment, then execute your Python script with arguments (if any) that you may need to pass to it.*

8. Create the Figcow submission script

- Create a file named **test_venv.sub** as below:

```
Universe   = vanilla
Executable = run_py.sh
Arguments  = /mnt/data/test_venv.py Hey Sunway

Output     = test_venv.out
Error      = test_venv.err
Log        = test_venv.log

Request_Cpus   = 1
Request_Memory = 2GB

Queue
```

9. Test your run_py script

- In your my_venv: **./run_py.sh test_venv.py hello world**

```
(my_venv) ubuntu@ip-172-31-92-248:~$ ./run_py.sh test_venv.py hello world
Hello, ubuntu!
You're running on the host ip-172-31-92-248
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(my_venv) ubuntu@ip-172-31-92-248:~$
```

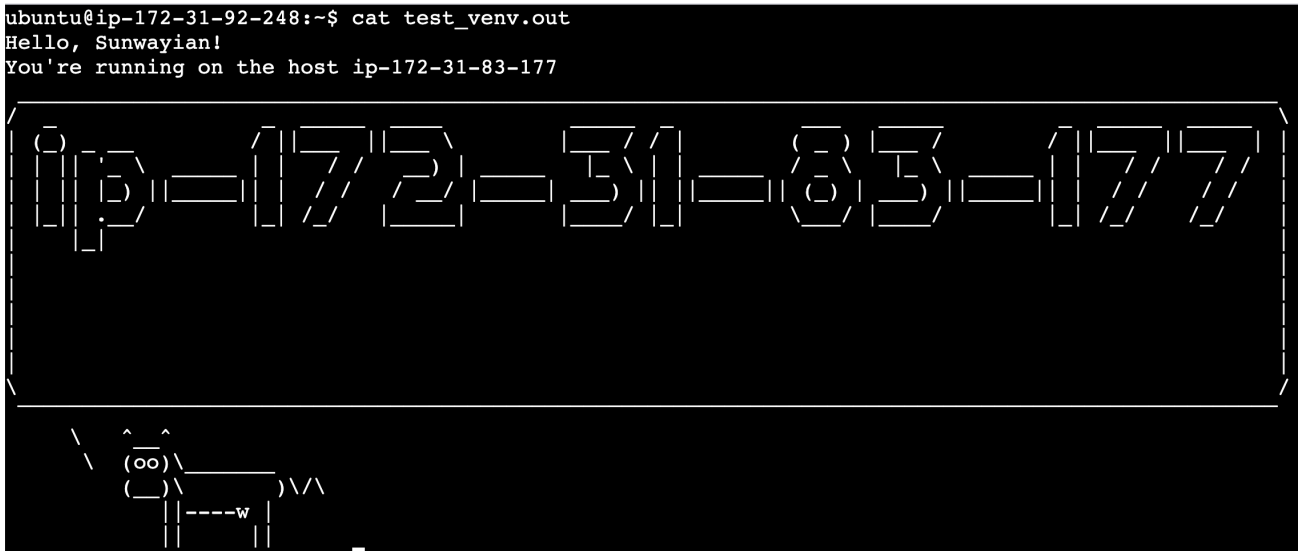
10. Submit your Figcow to Condor

- Submit to condor pool: **condor_submit test_venv.sub**

```
ubuntu@ip-172-31-92-248:~$ condor_submit test_venv.sub
Submitting job(s).
1 job(s) submitted to cluster 11.
ubuntu@ip-172-31-92-248:~$ cat test_venv.out
Hello, nobody!
You're running on the host ip-172-31-83-177
 _____________________________________________________________________
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ubuntu@ip-172-31-92-248:~$
```

Lab Test 3 Challenge!!!

- Modify your scripts, to show the hostname.



Submission to eLearn

Show your account

Show the private IPs

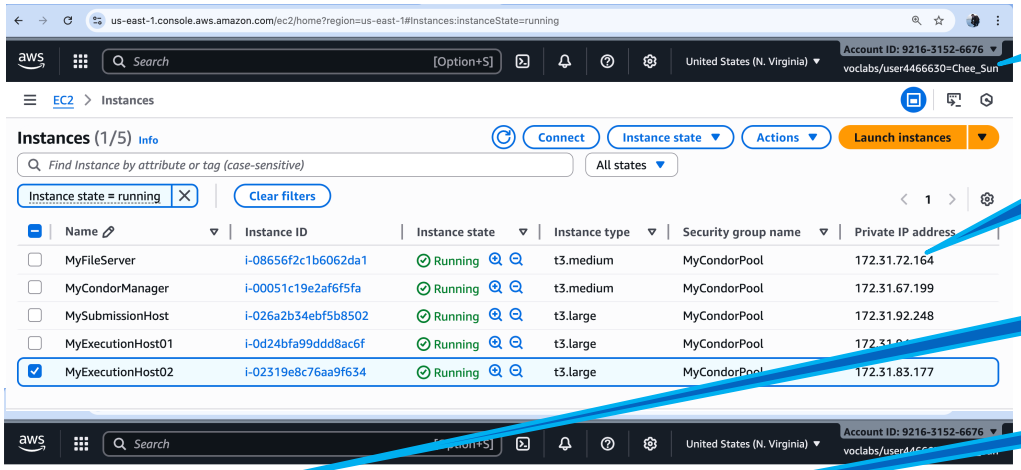
Show mounting point

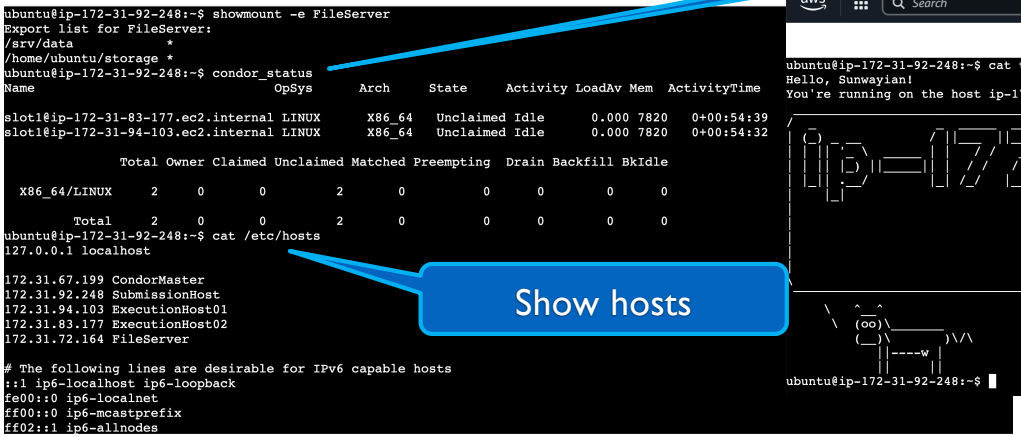
Show condor_status

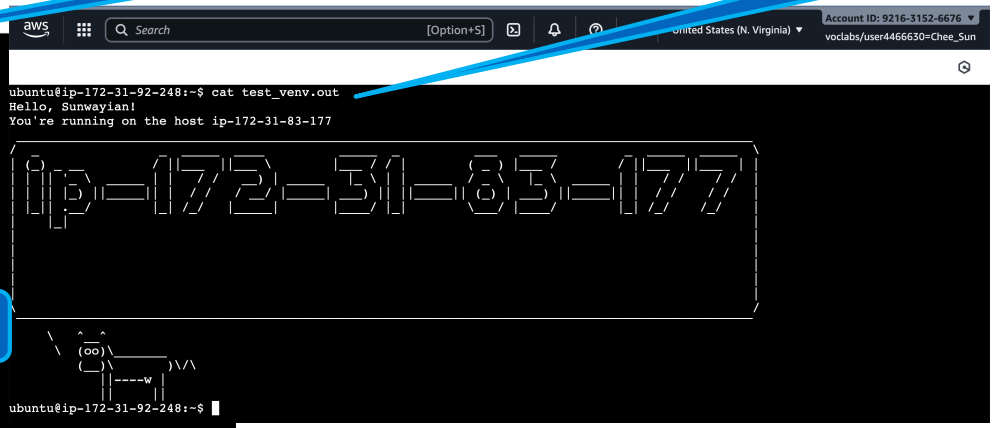
Show your output

Show hosts

1. Condor Pool: 4 marks
2. Show Figcow on Condor Pool: 6 marks







THANK YOU

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